

Azure vs. Active Directory – Student Technician Guide

1. Overview

UNC Charlotte is transitioning podium computers from **Active Directory (AD)** to **Azure/Entra** management. During this transition, student technicians will encounter **both systems** across campus. This guide explains how to identify each system, how login credentials differ, and what troubleshooting steps apply to each environment. Academic buildings will have both types of computers.

2. Login Identification & Technician Call Script

How to Identify the Login Method

On every NinerNet PC login screen, the **first box** tells you which credential type the computer requires:

- If the first box says “**Email Address**” → This is an **Azure** machine.
 - Login with **full @charlotte.edu email**
- If the first box says “**Username**” → This is an **Active Directory (AD)** machine.
 - Login with **NinerNet username only** (no “@charlotte.edu”)

Technician Call Script

Use this script verbatim when assisting callers:

“Thank you for calling Classroom Support. As the university transitions to a new computer management system, some campus computers will still use the old login method, while others now use the new login method. I can walk you through how to easily tell the difference.”

“On the login screen, please look at the **first box**. That box tells you which credentials you should use on that specific computer.”

“If the box says **Email Address**, please enter your full **@charlotte.edu** email.”

“If the box says **Username**, please enter just your **NinerNet username** — not the full email.”

“After entering the correct credential type, please enter your NinerNet password.”

3. System Differences (Student Tech Knowledge)

A. Basic Azure vs. AD Identification

Student techs should understand the following:

- **Standard Podium Experience (AD)**
 - Login: **NinerNet USERNAME**
 - Often uses **Dell Optimizer** for audio
 - BitLocker keys may be stored in AD or require escalation
- **New Podium Experience (Azure)**
 - Login: **NinerNet EMAIL**
 - Uses **Soundblaster** audio interface
 - Azure Portal tools available (restart, BitLocker keys, device info)
 - If it is a user's first time logging in to an Azure computer they will not be able to access their networked drives. They will have to sign out of the computer and then sign back in to access their networked drives.

4. Required Knowledge for Student Technicians

A. BitLocker Key Identification (Azure Machines)

- What is it?
 - A blue recovery prompt on windows machines that appears when the system detects a potential security risk or hardware change.
- What to do?
 - If you come across a podium machine that is asking for a bitlocker key please contact full time staff. Ask for the 48 digit numerical recovery code.
- How to document the incident in the ticket

B. Azure Restart Shortcut

- To restart an Azure computer without physically unplugging power use the **Azure Portal restart command**
 - This is an icon on the home page of the computer. It has a UNC Charlotte Logo and is labeled restart computer.

C. Soundblaster Troubleshooting

Azure rooms use **Soundblasters** instead of Dell Optimizer. A known problem with soundblasters is that they can freeze in the middle of use. Which can cause line-in audio in zoom or a recording to not work.

You may encounter an issue when a professor states mid-lecture that suddenly zoom does not detect a microphone, when previously it was working fine, or zoom may still see a microphone, but audio levels are not observable in zoom.

It's noteworthy this resolution only applies when the microphones are actually paired, and still outputting noise in the room. If the microphones are not outputting in the room, the mics might not be turned on, could be muted, or not working otherwise.

The soundblaster USB cards have dedicated sound out and sound in 3.5mm ports, and the AV signal flow to the PC in many rooms will follow:

- Wireless Microphone → Receiver → AVS → Soundblaster → PC

And

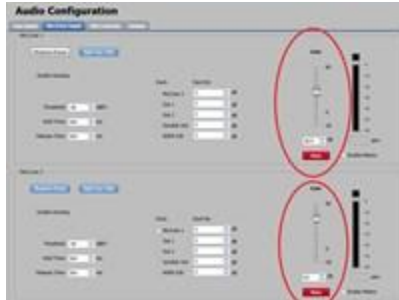
- Gooseneck Mic → AVS → Soundblaster → PC

The soundblaster crashing can happen 3 different ways:

1. **A Faulty Microphone (more likely to be the gooseneck) is sending loud crackles through the signal flow, causing the soundblaster to crash.**
2. **Incorrect levels on a microphone (gain too high), causing the soundblaster to crash.**
3. **Correct levels on a microphone, but the end user is shouting into the microphone, causing the soundblaster to crash.**

Way 1 can be easily verified by replacing the suspected faulty microphone, and then disconnecting and reconnecting the soundblaster to restart it.

Way 2 can be remedied by decreasing the individual microphone's gain, and then disconnecting and reconnecting the soundblaster to restart it. Ensure on the meter that it is not peaking when speaking at a normal level.



Way 3 can be remedied by decreasing the variable output dB on left and right, and then disconnecting and reconnecting the soundblaster to restart it.



D. Quarantine Identification

This applies to Azure and AD machines

Azure devices may enter **quarantine** if:

- Compliance checks fail
- Network policies trigger isolation
- Security flags are raised

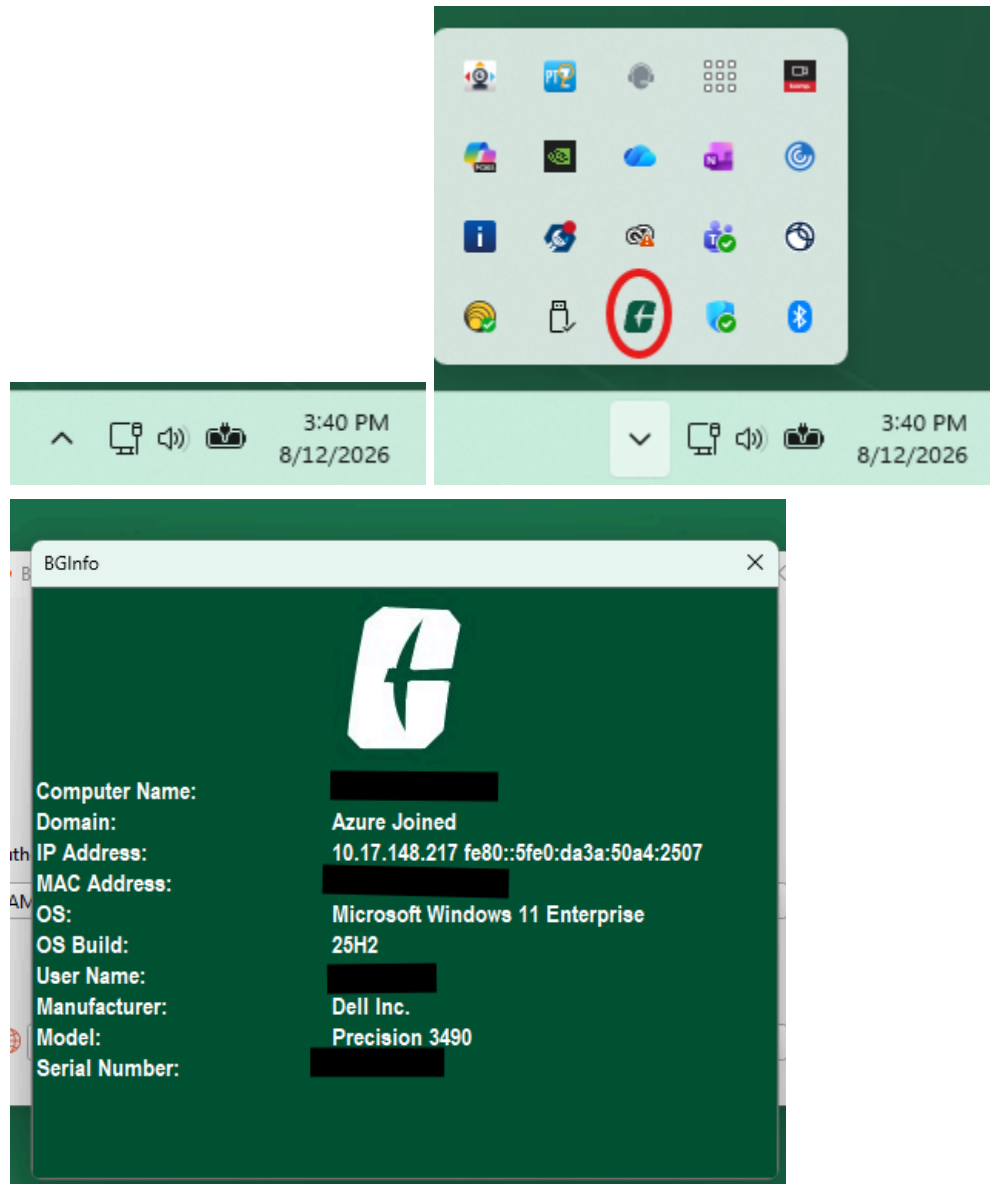
Symptoms:

- The computer might show that it is not connected to the network
- The computer could show it is connected to the network and the user was able to login but university websites like canvas will not load.

Identifying:

- If you think a computer is in quarantine. The way to verify is by checking the IP address. A NinerNet PC's should be a 10.18, 10.17, 10.16.
 - **A Quarantined IP is 10.41 or 10.41**
- There are two ways you can check a computer's IP address.

- The first is by using the menu bar. You click on the upwards marker in the taskbar and then on the UNC Charlotte logo. The pictures below show the steps.



- The second way is by opening the command prompt, which is an app on the computer you can search for in the task bar. Then type in ipconfig. It will load multiple lines of information. You are looking for "IPV4 Address:"

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Command Prompt

Ethernet adapter Ethernet 2:

    Connection-specific DNS Suffix  . : dyn.uncc.edu
    Link-local IPv6 Address . . . . . : fe80::5fe0:da3a:50a4:2507%3
    IPv4 Address. . . . . : 10.17.148.217
    Subnet Mask . . . . . : 255.255.254.0
    Default Gateway . . . . . : 10.17.148.1

Wireless LAN adapter Local Area Connection* 1:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 2:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : uncc.edu

Ethernet adapter Bluetooth Network Connection:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

C:\Users\ebrowl80>
```

Trouble shooting steps:

- If you think a computer could be in quarantine or it is showing that it is disconnected from the network the first thing you are going to do is to check the Ip address.
- If the IP address is 10.40 or 10.41 please notify full time staff that that specific computer is in quarantine and then complete the steps below while full time staff contacts the universities networking team.
 - Open the back of the podium and physically disconnect the power cable and the network cable from the NinerNet PC. Leave it disconnected for at least 30 seconds. Then reconnect it. This is to allow that device to disconnect from the network completely, so that when it is reconnected it will authenticate correctly.
 - This is not a guaranteed solution so please let full time staff know that the computer is in quarantine so we can start the escalation process to networking.
 - If the restart does not work, the fastest solution for a professor or end user is to use a laptop through a cable cubby connection.

We have seen an increase in the amount of quarantine machines over this summer. We understand that it is frustrating to be in a situation where the solution to a problem is not necessarily immediately available. Our job in this situation is to do our due diligence and communicate the issue and the action plan to the professors/end users calmly and respectfully.

E. Guest Account Limitations

Guest accounts behave differently on Azure vs. AD:

- On AD Computers users can use the presenter/presents account
- On Azure computers users can click the guest account button in the corner. No information will be saved on this guest account.